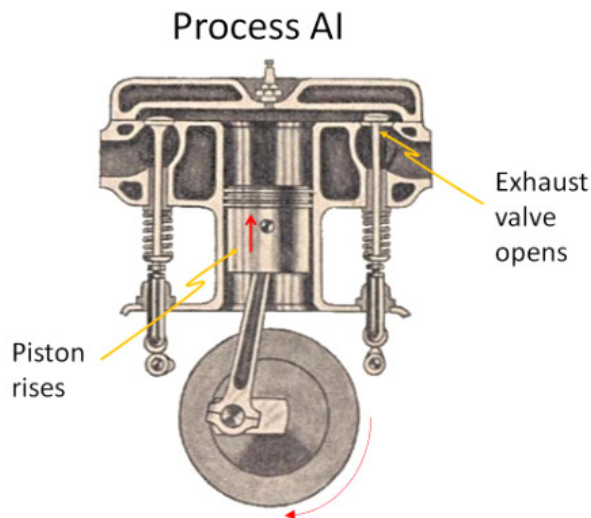


## 1.5 Process $A \rightarrow I$ and process $I \rightarrow A$

Assignment:

Fill out the following table for process  $A \rightarrow I$

| State   | T (K) | P (kPa)              | V (m <sup>3</sup> )   | n (mol)                 | E <sub>int</sub> (J) | —              |
|---------|-------|----------------------|-----------------------|-------------------------|----------------------|----------------|
| A       | 293   | 100                  | 0.000 5               | $2.0525 \times 10^{-2}$ | 125.00               | —              |
| I       | 293   | 100                  | $6.25 \times 10^{-5}$ |                         |                      | —              |
| Process | Q (J) | w <sub>int</sub> (J) | ΔE <sub>int</sub> (J) | w <sub>eng</sub> (J)    | Q <sub>h</sub>       | Q <sub>c</sub> |
| AI      |       |                      |                       |                         |                      |                |



But we are not done. The opening of the valve let the spent fuel begin to leave as exhaust. But we can help remove the exhaust by pushing the piston up. This is done at constant pressure, because the valve is open to the exhaust system air pressure. So this is almost an isobaric process. But it's not quite isobaric because we will lose gas by venting it. The number of moles,  $n$ , will change. We can find the new number of moles using the ideal gas law

$$P_I V_I = nRT$$

since  $P_I = P_A$  and  $V_I = V_B$  because the piston minimum volume is the same each time the piston goes up. Then we can use our compression ratio for  $V_A$  and  $V_I$

$$\frac{V_A}{V_I} = 8.00$$

or, solving for  $V_I$

$$V_I = 8/V_A$$

which we could calculate, but it is just the same as  $V_B$

We want  $n_I$ , so let's use the ideal gas law for points  $A$  and  $I$ . We get

$$P_A V_A = n_A R T_A$$

and

$$P_I V_I = n_I R T_I$$

We can divide these two equations

$$\frac{P_A V_A}{P_I V_I} = \frac{n_A R T_A}{n_I R T_I}$$

and note that we have our compression ratio hidden in the left hand side, and recall that  $P_I = P_A$  so the left hand side is just 8. And on the right hand side  $T_A = T_I$  because we didn't change the temperature of the gas, we just moved it.

$$8 = \frac{n_A}{n_I}$$

We can solve this for the number of moles at piston  $I$

$$n_I = \frac{n_A}{8}$$

And we have all the numbers, so

$$\begin{aligned} n_I &= \frac{(2.0525 \times 10^{-2} \text{ mol})}{8} \\ &= 2.5656 \times 10^{-3} \text{ mol} \end{aligned}$$

And then  $E_I$  will be

$$\begin{aligned} E_D &= \frac{5}{2} n_I R T_I \\ &= \frac{5}{2} (2.5656 \times 10^{-3} \text{ mol}) \left( 8.314 \frac{\text{J}}{\text{mol K}} \right) (293 \text{ K}) \\ &= 15.625 \text{ J} \end{aligned}$$

The work done for a true isobaric process is

$$w_{AI} = P_A (V_I - V_A)$$

but this assumed that  $n$  did not change. Really we are just moving the gas, and it doesn't take much of a force to move the little bit of gas. So to a good approximation we can say

$$\begin{aligned} w_{int} &= F \Delta x \\ &\approx 0 \end{aligned}$$

and we can ignore  $w_{AI}$ .

For finding  $Q_{AI}$  we would like to use

$$Q = nC_P\Delta T$$

but we quickly run into trouble again because  $n$  changed! We are using convection to remove heat instead of conduction. We can get around this by finding  $\Delta E_{int}$  directly.

$$\begin{aligned}\Delta E_{AI} &= E_I - E_A \\ &= 15.625 \text{ J} - 125.00 \text{ J} \\ &= -109.38 \text{ J}\end{aligned}$$

From this and the first law

$$\Delta E_{int} = Q + w_{int}$$

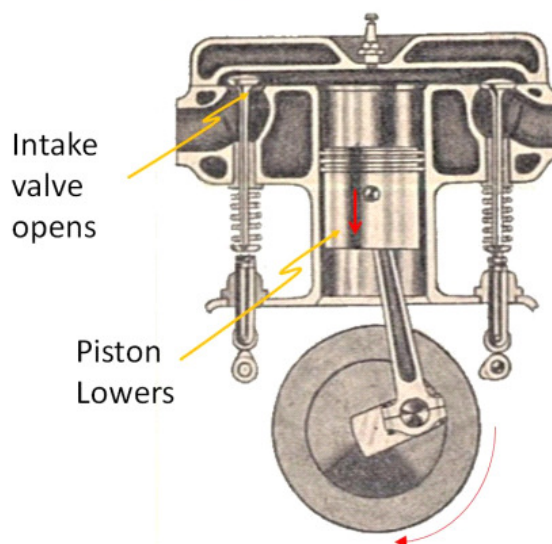
We can find  $Q$

$$\begin{aligned}Q &\approx \Delta E_{int} \\ &\approx -109.38 \text{ J}\end{aligned}$$

Energy is leaving with the gas, so we seem to pick up a contribution to  $Q_c$ . But really this is not part of our thermodynamic process, because we mechanically removed the energy by removing the gas. Let's denote this by putting our energy in parenthesis to remind us that this is a more mechanical than thermal process.

| State     | T (K)       | P (kPa)              | V (m <sup>3</sup> )   | n (mol)                   | E <sub>int</sub> (J) | —              |
|-----------|-------------|----------------------|-----------------------|---------------------------|----------------------|----------------|
| <b>A</b>  | 293         | 100                  | 0.000 5               | $2.052\,5 \times 10^{-2}$ | 125.00               | —              |
| <b>I</b>  | 293         | 100                  | $6.25 \times 10^{-5}$ | $2.565\,6 \times 10^{-3}$ | 15.625 J             | —              |
| Process   | Q (J)       | w <sub>int</sub> (J) | ΔE <sub>int</sub> (J) | w <sub>eng</sub> (J)      | Q <sub>h</sub>       | Q <sub>c</sub> |
| <b>AI</b> | (−109.38 J) | ~ 0                  | (−109.38)             | ~ 0                       | 0                    | (109.38)       |

## Process IA



Now the exhaust valve closes, and the intake valve opens. The new fuel mixture is brought in. We essentially reverse process *AI* in process *IA*.

| State     | $T$ (K)    | $P$ (kPa)     | $V$ ( $\text{m}^3$ )  | $n$ (mol)               | $E_{int}$ (J) | —     |
|-----------|------------|---------------|-----------------------|-------------------------|---------------|-------|
| <b>I</b>  | 293        | 100           | $6.25 \times 10^{-5}$ | $2.5657 \times 10^{-3}$ | 15.625 J      | —     |
| <b>A</b>  | 293        | 100           | 0.000 5               | $2.0525 \times 10^{-2}$ | 125.00        | —     |
| Process   | $Q$ (J)    | $w_{int}$ (J) | $\Delta E_{int}$ (J)  | $w_{eng}$ (J)           | $Q_h$         | $Q_c$ |
| <b>IA</b> | (109.38 J) | $\sim 0$      | (109.38 J)            | $\sim 0$                | (−109.38 J)   | 0     |

But there is a difference. The new gas has useful fuel. and that means we are ready to complete the cycle again by compressing the gas, igniting it, and receiving the power stroke and exhaust. Note that Processes *AI* and *IA* have a net zero effect thermodynamically, but are very important to making your car or lawn mower go!